



Aurora Sail

Tri-Boom Solar Sail Mechanism

The Aurora Sail is a precision-engineered deployment mechanism for high-area CubeSat solar sails. It utilizes a refined modular Tri-Boom architecture to maximize packing efficiency.

By integrating precision carbon fiber stabilizers, the system provides a low-friction, high-reliability solution for deploying large-scale membranes in complex orbital environments.

SPECIFICATIONS

Configuration	Tri-Boom
Boom Material	Composite Slit-Tube
Stowed Volume	~80mm Dia (Tuna Can)
Sail Area	0.425 m ² (Scalable)
Sail Material	CP1 Polyimide
Mass (System)	140g (excl. motor)
Actuation	NEMA 17 / BLDC
TRL	4 (Validation)

KEY FEATURES

Architecture	Modular Tri-Boom
Guide Rails	3mm Carbon Fiber
Actuation	Burn Wire Release
Feedback	Limit Switch